THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 7

Groups: The Language of Symmetry

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. The four axioms of a group

(write them in your own words, not the formal versions)

____

____

2. What a group captures conceptually

"Reversible ____________ that can be ____________"

____

____

3. Three examples of groups from different domains

a.

b.

c.

____

____

4. A homomorphism preserves

An isomorphism is:
"Same structure, different ____________"

5. The organizing principle

"The right maps between structures are those that ____________"

6. Subgroups let you

Quotient groups let you:
Together they let you:

7. A representation is

Why representations matter for applications:

8. The connection between groups and Chapter 5 (symmetry)

how does the formal definition capture the informal idea?